

Reviews

Reviewer #1:

- The paper needs an additional part to **compare the original SSC parameters with the ESSC parameters in their performances**, **SE JUSTIFICA CON EL FEATURE RANKING**
- The authores need to show why the performances are lowered although SNR still High.

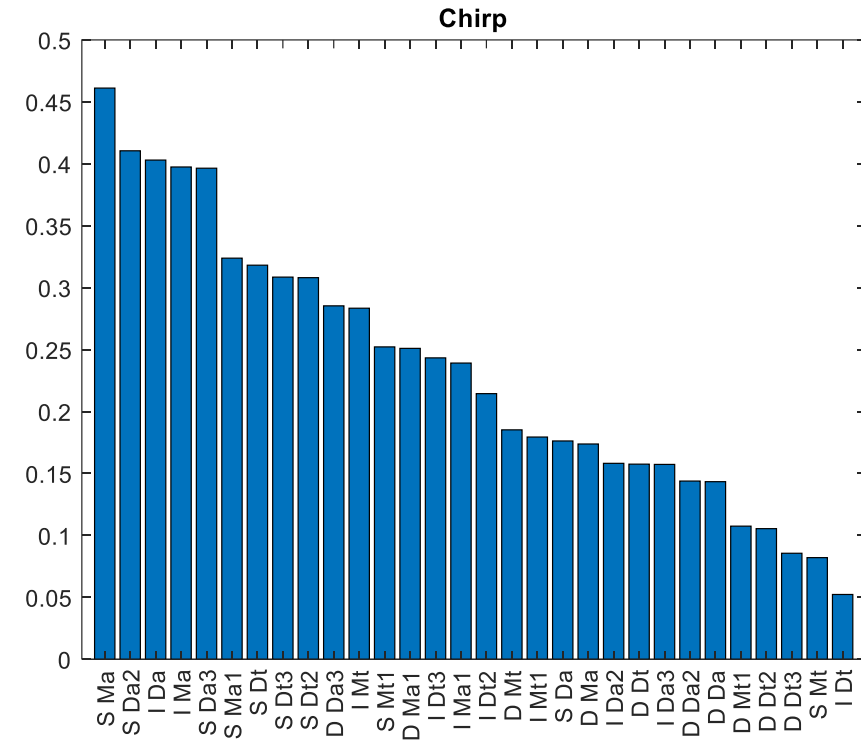
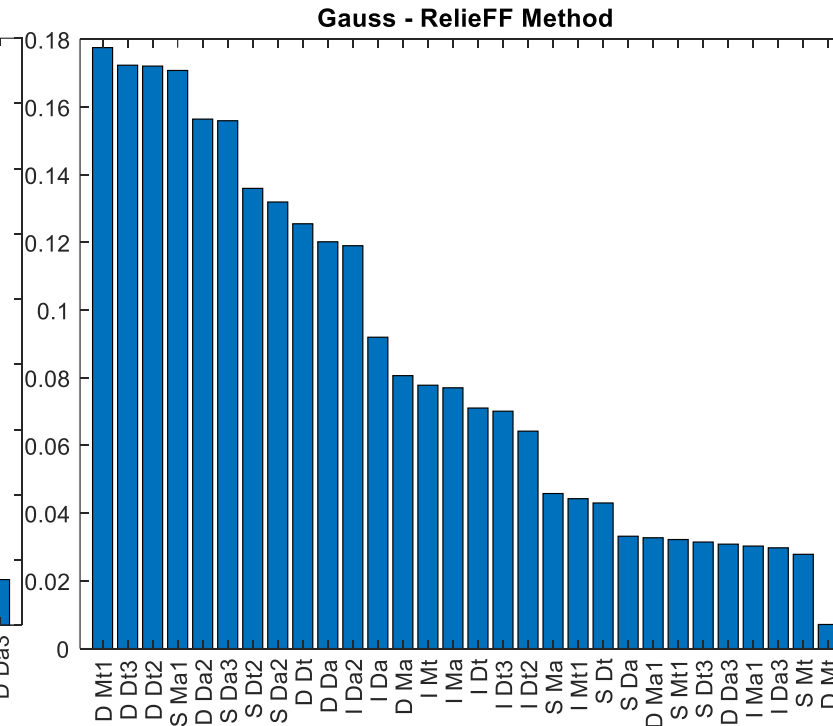
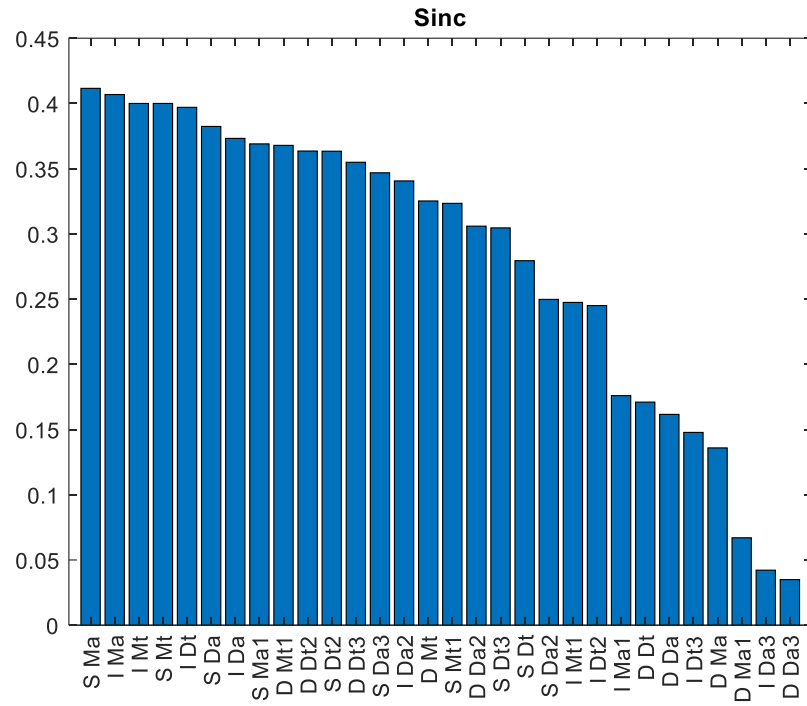
Reviewer #2:

- The manuscript would benefit from a clearer, more verifiable articulation of novelty. In its current form, ESSC appears to combine multiple statistical descriptors (including derivative and integral transforms), but the **design choices** (e.g., 30 features, truncation at third order) are not sufficiently justified. Please add an explicit contributions list and **provide a principled rationale for feature selection and truncation**, as well as expected applicability limits.
- The preprocessing pipeline involves multiple fixed hyperparameters (window sizes, thresholds), yet robustness across sampling rates, pulse widths, noise types, and N_s is not demonstrated. **Please include a sensitivity analysis** (e.g., $\pm 50\%$ perturbations) reporting accuracy and feature stability or propose adaptive parameter-setting rules tied to N_s , estimated pulse width, or noise statistics.
- **Feature Selection, Feature Ranking Algorithms.** Feature importance results based on ReliefF are correlational; they do not establish necessity. **Please provide ablations by signal source** (f, f_D, f_I and combinations) and by feature groups (SSC vs time-distribution vs amplitude-distribution statistics), and report performance changes with confidence intervals across multiple runs.

Reviews

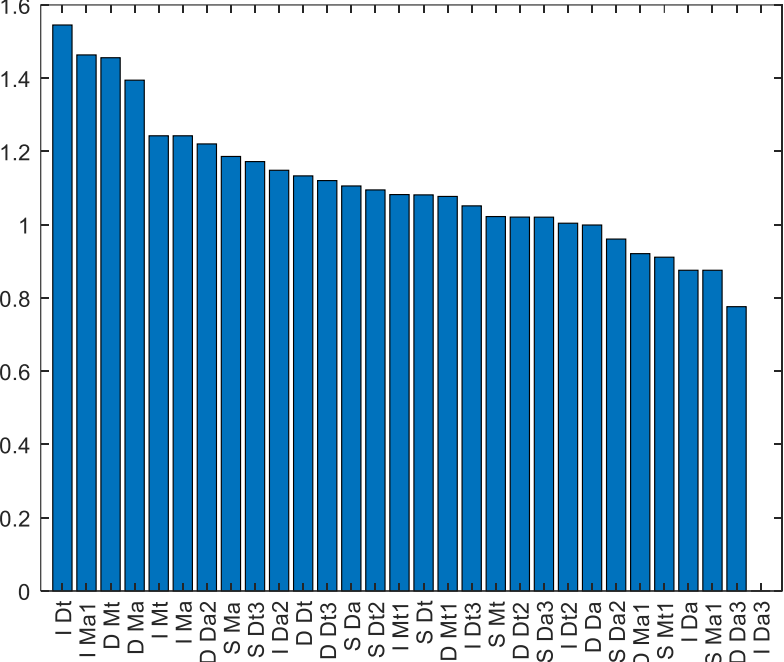
- Cómo justificamos la degradación de los clasificadores a al disminuir el valor de SNR. Es definir las limitaciones del método.
- Recordar todo el proceso de filtrado y detección, el filtrado deforma la señal corriendola de las deformaciones con la cual el modelo fue entrenado.
- El método no es immune al ruido. Así como también no es usual encontrar señales en las aplicaciones de RMN con este nivel de ruido.
- Recordar aquí que los subsets son una mezcla de parámetros.
- El método tiene limitaciones muy grandes ante ruido, el proceso de filtrado y detección de pulso, obtiene los params ESSC de un pulso deformado que se corre demasiado del tipo de señal deformada para la cual fue entrenado.
- Qué método da un major ranking? No obstante, es muy difícil clasificar deformaciones en señales gaussianas, dado que los tipos de filtros provocan deformaciones muy sutiles, la extension de parámetros mejora a la MLP.

Rankeo Método RELIEFF

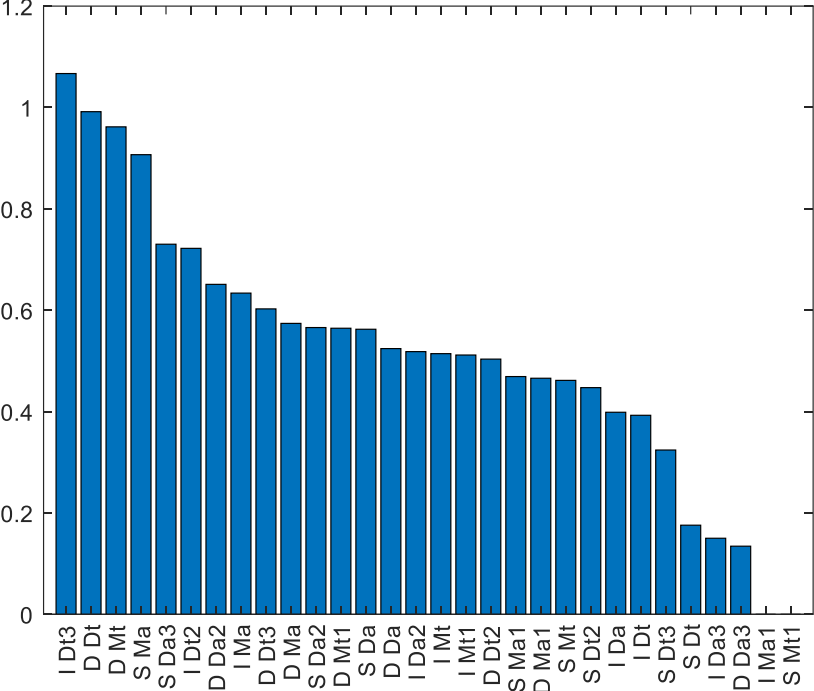


Rankeo Método MRMR

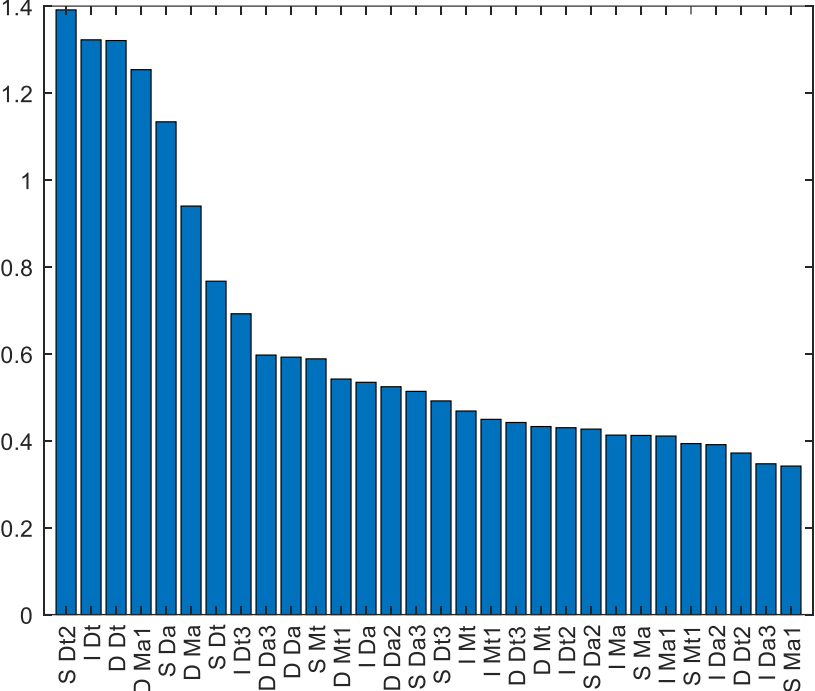
Sinc



Gauss



Chirp



Tablas con los subsets (SINC)

- SUBSETS SEGÚN EL MÉTODO DE RANKEO**

SINC Rank ReliefF

$\mathcal{S}_{(5)}^1$	S Ma'	'I Ma'	'I Mt'	'S Mt'	'I Dt'															
$\mathcal{S}_{(10)}^2$	S Ma'	'I Ma'	'I Mt'	'S Mt'	'I Dt'	'S Da'	'I Da'	'S Ma1'	'D Mt1'	'D Dt2'										
$\mathcal{S}_{(15)}^3$	S Ma'	'I Ma'	'I Mt'	'S Mt'	'I Dt'	'S Da'	'I Da'	'S Ma1'	'D Mt1'	'D Dt2'	'S Dt2'	'D Dt3'	'S Da3'	'I Da2'	'D Mt'					
$\mathcal{S}_{(20)}^4$	S Ma'	'I Ma'	'I Mt'	'S Mt'	'I Dt'	'S Da'	'I Da'	'S Ma1'	'D Mt1'	'D Dt2'	'S Dt2'	'D Dt3'	'S Da3'	'I Da2'	'D Mt'	'S Mt1'	'D Da2'	'S Dt3'	'S Dt'	'S Da2'

Sinc Rank MRMR

$S^1_{(5)}$	'I Dt'	'I Ma1'	'D Mt'	'D Ma'	'I Mt'															
$S^2_{(10)}$	'I Dt'	'I Ma1'	'D Mt'	'D Ma'	'I Mt'	'I Ma'	'D Da2'	'S Ma'	'S Dt3'	'I Da2'										
$S^3_{(15)}$	'I Dt'	'I Ma1'	'D Mt'	'D Ma'	'I Mt'	'I Ma'	'D Da2'	'S Ma'	'S Dt3'	'I Da2'	'D Dt'	'D Dt3'	'S Da'	'S Dt2'	'I Mt1'					
$S^4_{(20)}$	'I Dt'	'I Ma1'	'D Mt'	'D Ma'	'I Mt'	'I Ma'	'D Da2'	'S Ma'	'S Dt3'	'I Da2'	'D Dt'	'D Dt3'	'S Da'	'S Dt2'	'I Mt1'	'S Dt'	'D Mt1'	'I Dt3'	'S Mt'	'D Dt2'

SUBSETS SEGÚN EL MÉTODO DE RANKEO

GAUSS Rank ReliefF

$S^1_{(5)}$	'D Mt1'	'D Dt3'	'D Dt2'	'S Ma1'	'D Da2'															
$S^2_{(10)}$	'D Mt1'	'D Dt3'	'D Dt2'	'S Ma1'	'D Da2'	'S Da3'	'S Dt2'	'S Da2'	'D Dt'	'D Da'										
$S^3_{(15)}$	'D Mt1'	'D Dt3'	'D Dt2'	'S Ma1'	'D Da2'	'S Da3'	'S Dt2'	'S Da2'	'D Dt'	'D Da'	'l Da2'	'l Da'	'D Ma'	'l Mt'	'l Ma'					
$S^4_{(20)}$	'D Mt1'	'D Dt3'	'D Dt2'	'S Ma1'	'D Da2'	'S Da3'	'S Dt2'	'S Da2'	'D Dt'	'D Da'	'l Da2'	'l Da'	'D Ma'	'l Mt'	'l Ma'	'l Dt'	'l Dt3'	'l Dt2'	'S Ma'	'l Mt1'

GAUSS Rank MRMR

$S^1_{(5)}$	'I Dt3'	'D Dt'	'D Mt'	'S Ma'	'S Da3'															
$S^2_{(10)}$	'I Dt3'	'D Dt'	'D Mt'	'S Ma'	'S Da3'	'I Dt2'	'D Da2'	'I Ma'	'D Dt3'	'D Ma'										
$S^3_{(15)}$	'I Dt3'	'D Dt'	'D Mt'	'S Ma'	'S Da3'	'I Dt2'	'D Da2'	'I Ma'	'D Dt3'	'D Ma'	'S Da2'	'D Mt1'	'S Da'	'D Da'	'I Da2'					
$S^4_{(20)}$	'I Dt3'	'D Dt'	'D Mt'	'S Ma'	'S Da3'	'I Dt2'	'D Da2'	'I Ma'	'D Dt3'	'D Ma'	'S Da2'	'D Mt1'	'S Da'	'D Da'	'I Da2'	'I Mt'	'I Mt1'	'D Dt2'	'S Ma1'	'D Ma1'

Tablas con los subsets (CHIRP)

- SUBSETS SEGÚN EL MÉTODO DE RANKEO**

CHIRP Rank ReliefF

$S^1_{(5)}$	'S Ma'	'S Da2'	'I Da'	'I Ma'	'S Da3'															
$S^2_{(10)}$	'S Ma'	'S Da2'	'I Da'	'I Ma'	'S Da3'	'S Ma1'	'S Dt'	'S Dt3'	'S Dt2'	'D Da3'										
$S^3_{(15)}$	'S Ma'	'S Da2'	'I Da'	'I Ma'	'S Da3'	'S Ma1'	'S Dt'	'S Dt3'	'S Dt2'	'D Da3'	'I Mt'	'S Mt1'	'D Ma1'	'I Dt3'	'I Ma1'					
$S^4_{(20)}$	'S Ma'	'S Da2'	'I Da'	'I Ma'	'S Da3'	'S Ma1'	'S Dt'	'S Dt3'	'S Dt2'	'D Da3'	'I Mt'	'S Mt1'	'D Ma1'	'I Dt3'	'I Ma1'	'I Dt2'	'D Mt'	'I Mt1'	'S Da'	'D Ma'

CHIRP Rank MRMR

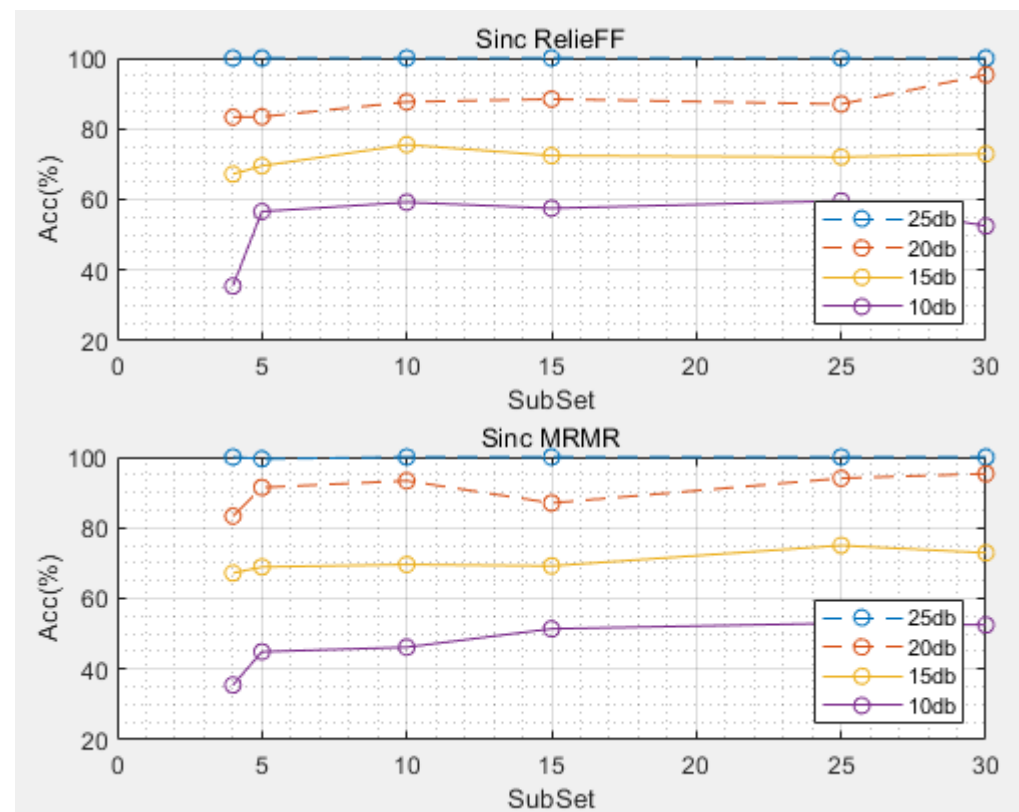
$S^1_{(5)}$	'S Dt2'	'I Dt'	'D Dt'	'D Ma1'	'S Da'															
$S^2_{(10)}$	'S Dt2'	'I Dt'	'D Dt'	'D Ma1'	'S Da'	'D Ma'	'S Dt'	'I Dt3'	'D Da3'	'D Da'										
$S^3_{(15)}$	'S Dt2'	'I Dt'	'D Dt'	'D Ma1'	'S Da'	'D Ma'	'S Dt'	'I Dt3'	'D Da3'	'D Da'	'S Mt'	'D Mt1'	'I Da'	'D Da2'	'S Da3'					
$S^4_{(20)}$	'S Dt2'	'I Dt'	'D Dt'	'D Ma1'	'S Da'	'D Ma'	'S Dt'	'I Dt3'	'D Da3'	'D Da'	'S Mt'	'D Mt1'	'I Da'	'D Da2'	'S Da3'	'S Dt3'	'I Mt'	'I Mt1'	'D Dt3'	'D Mt'

ACC% de cada subset y SNR

Cómo se comportan los modelos entrenados con los subsets según los niveles de SNR.

----- SINC -----						
Relieff Sobre el MLP						
	Params Number / Acc% /					
SNR	4(SSC)	5	10	15	20	30(ESSC)
25db	100	100	100	100	100	100
20db	83.3	83.4	87.6	88.4	87	95.3
15db	67.2	69.5	75.5	72.4	72	72.9
10db	35.5	56.6	59.2	57.5	59.6	52.5

----- SINC -----						
MRMR Sobre el MLP						
	Params Number / Acc% /					
SNR	4(SSC)	5	10	15	20	30(ESSC)
25db	100	99.5	100	100	100	100
20db	83.3	91.4	93.3	87	94	95.3
15db	67.2	68.9	69.6	69.2	75	72.9
10db	35.5	44.9	46.2	51.4	53	52.5

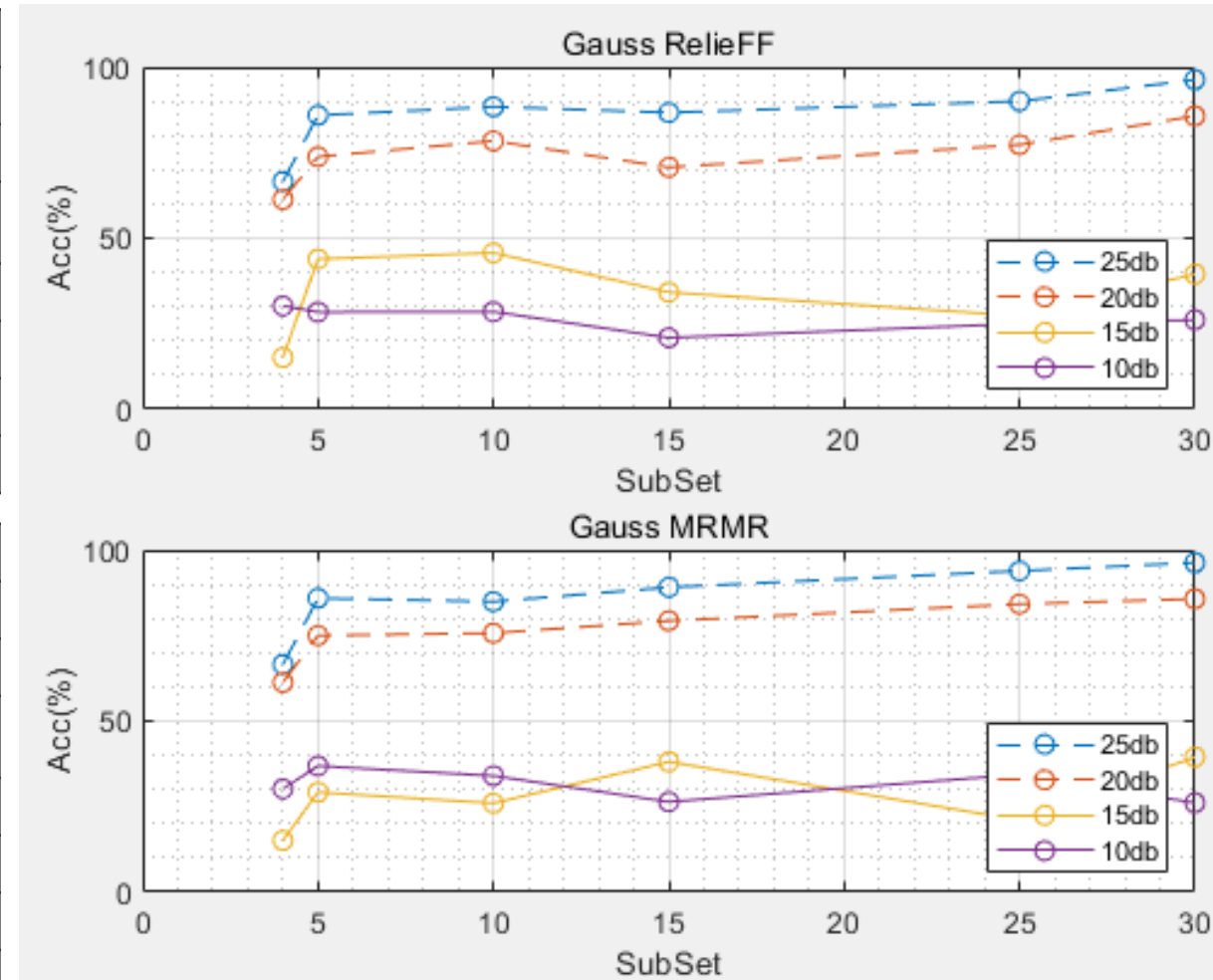


ACC% de cada subset y SNR

Cómo se comportan los modelos entrenados con los subsets según los niveles de SNR.

----- GAUSS -----						
Relieff Sobre el MLP						
	Params Number / Acc% /					
SNR	4(SSC)	5	10	15	20	30(ESSC)
25db	66.5	85.9	88.4	86.7	90	96.3
20db	61.3	73.8	78.4	70.7	77.3	85.7
15db	15.1	43.9	45.7	34.2	27.3	39.5
10db	30.2	28.4	28.5	20.9	25	26.1

----- GAUSS -----						
MRMR Sobre el MLP						
	Params Number / Acc% /					
SNR	4(SSC)	5	10	15	20	30(ESSC)
25db	66.5	85.9	84.9	89.1	93.9	96.3
20db	61.3	75	75.7	79.3	84.2	85.7
15db	15.1	29.2	26.0	38.1	21.1	39.5
10db	30.2	36.9	34	26.4	34.4	26.1

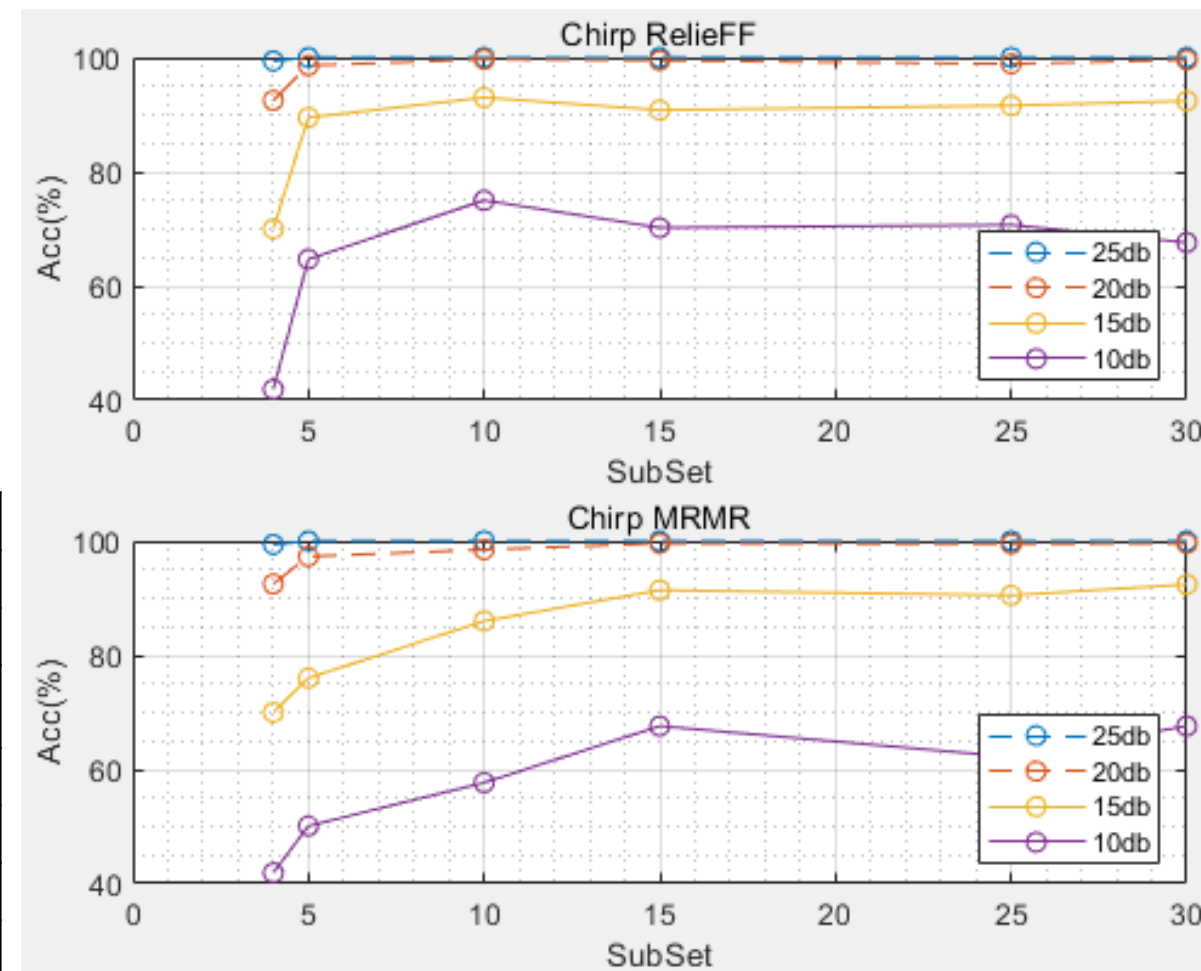


ACC% de cada subset y SNR

Cómo se comportan los modelos entrenados con los subsets según los niveles de SNR.

----- CHIRP -----						
Relieff Sobre el MLP						
	Params Number / Acc% /					
SNR	4(SSC)	5	10	15	20	30(ESSC)
25db	99.4	100	100	100	100	100
20db	92.5	98.6	99.7	99.5	98.9	99.6
15db	70	89.5	93	90.8	91.6	92.4
10db	41.9	64.7	75	70.2	70.7	67.6

----- CHIRP -----						
MRMR Sobre el MLP						
	Params Number / Acc% /					
SNR	4(SSC)	5	10	15	20	30(ESSC)
25db	99.4	100	100	100	100	100
20db	92.5	97.3	98.5	99.6	99.5	99.6
15db	70	76	86.0	91.4	90.5	92.4
10db	41.9	50.1	57.7	67.6	62.2	67.6



Parámetros de los modelos 30params vs 4 scc

Sinc Sinc 4 params

Model 5: Optimizable Neural Network
Status: Trained

Training Results

Accuracy (Validation) 100.0%
Total cost (Validation) 0
Prediction speed ~120000 obs/sec
Training time 702.05 sec
Model size (Compact) ~16 kB

Model Hyperparameters

Preset: Optimizable Neural Network
Iteration limit: 1000
Standardize data: Yes

Optimized Hyperparameters

Number of fully connected layers: 2
Activation: Tanh
Regularization strength (Lambda): 3.2134e-05
First layer size: 156
Second layer size: 3

Hyperparameter Search Range

Number of fully connected layers: 1-3
Activation: ReLU, Tanh, Sigmoid, None
Regularization strength (Lambda): 2e-09-20
First layer size: 1-300
Second layer size: 1-300
Third layer size: 1-300

Sinc Sinc 30 params

Model 5: Optimizable Neural Network
Status: Trained

Training Results

Accuracy (Validation) 100.0%
Total cost (Validation) 0
Prediction speed ~120000 obs/sec
Training time 938.36 sec
Model size (Compact) ~66 kB

Model Hyperparameters

Preset: Optimizable Neural Network
Iteration limit: 1000
Standardize data: Yes

Optimized Hyperparameters

Number of fully connected layers: 1
Activation: Sigmoid
Regularization strength (Lambda): 9.6802e-06
First layer size: 202

Hyperparameter Search Range

Number of fully connected layers: 1-3
Activation: ReLU, Tanh, Sigmoid, None
Regularization strength (Lambda): 2e-09-20
First layer size: 1-300
Second layer size: 1-300
Third layer size: 1-300

Gauss Sinc 4 params

Model 5: Optimizable Neural Network
Status: Trained

Training Results

Accuracy (Validation) 66.1%
Total cost (Validation) 1696
Prediction speed ~99000 obs/sec
Training time 3401.6 sec
Model size (Compact) ~331 kB

Model Hyperparameters

Preset: Optimizable Neural Network
Iteration limit: 1000
Standardize data: Yes

Optimized Hyperparameters

Number of fully connected layers: 3
Activation: ReLU
Regularization strength (Lambda): 2.6565e-08
First layer size: 198
Second layer size: 141
Third layer size: 84

Hyperparameter Search Range

Number of fully connected layers: 1-3
Activation: ReLU, Tanh, Sigmoid, None
Regularization strength (Lambda): 2e-09-20
First layer size: 1-300
Second layer size: 1-300
Third layer size: 1-300

Gauss Sinc 30 params

Model 5: Optimizable Neural Network
Status: Trained

Training Results

Accuracy (Validation) 96.1%
Total cost (Validation) 193
Prediction speed ~64000 obs/sec
Training time 3432.6 sec
Model size (Compact) ~246 kB

Model Hyperparameters

Preset: Optimizable Neural Network
Iteration limit: 1000
Standardize data: Yes

Optimized Hyperparameters

Number of fully connected layers: 3
Activation: Tanh
Regularization strength (Lambda): 1.0145e-05
First layer size: 88
Second layer size: 294
Third layer size: 4

Hyperparameter Search Range

Number of fully connected layers: 1-3
Activation: ReLU, Tanh, Sigmoid, None
Regularization strength (Lambda): 2e-09-20
First layer size: 1-300
Second layer size: 1-300
Third layer size: 1-300

Chirp Sinc 4 params

Model 5: Optimizable Neural Network
Status: Trained

Training Results

Accuracy (Validation) 99.6%
Total cost (Validation) 22
Prediction speed ~72000 obs/sec
Training time 1777.7 sec
Model size (Compact) ~79 kB

Model Hyperparameters

Preset: Optimizable Neural Network
Iteration limit: 1000
Standardize data: Yes

Optimized Hyperparameters

Number of fully connected layers: 2
Activation: Sigmoid
Regularization strength (Lambda): 2.4585e-06
First layer size: 30
Second layer size: 253

Hyperparameter Search Range

Number of fully connected layers: 1-3
Activation: ReLU, Tanh, Sigmoid, None
Regularization strength (Lambda): 2e-09-20
First layer size: 1-300
Second layer size: 1-300
Third layer size: 1-300

Chirp Sinc 30 params

Model 5: Optimizable Neural Network
Status: Trained

Training Results

Accuracy (Validation) 100.0%
Total cost (Validation) 1
Prediction speed ~57000 obs/sec
Training time 428.93 sec
Model size (Compact) ~700 kB

Model Hyperparameters

Preset: Optimizable Neural Network
Iteration limit: 1000
Standardize data: Yes

Optimized Hyperparameters

Number of fully connected layers: 3
Activation: Tanh
Regularization strength (Lambda): 2.0467e-07
First layer size: 36
Second layer size: 253
Third layer size: 300

Hyperparameter Search Range

Number of fully connected layers: 1-3
Activation: ReLU, Tanh, Sigmoid, None
Regularization strength (Lambda): 2e-09-20
First layer size: 1-300
Second layer size: 1-300
Third layer size: 1-300